WORKED SOLUTION

STEP III 1988, Question 2

Pure • Linear Recurrence Relations

STEP III 1988, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s3-q2

The key idea

A linear recurrence with constant coefficients behaves like a quadratic: powers x^n solve it exactly when x is a root of the auxiliary equation. Since the recurrence fixes every term once the first two are known, matching u_0 and u_1 is all it takes to get the only solution. The last part has variable coefficients, but the factors $(n+1)$, $(n+2)$, $(n+3)$ suggest dividing through by their product, and then a simple substitution turns it into a constant-coefficient recurrence of the first kind.

Solution

(The paper prints the leading coefficient of the recurrence as α. This is clearly a misprint for a, matching the quadratic ax^2+bx+c, and we read it that way.) Since the quadratic has two roots, it really is a quadratic, so $a \neq 0$.

The formula satisfies the recurrence

Put $u_n = A\alpha^n + B\beta^n$ for any constants A, B. Then

$$\begin{aligned} au_{n+2} + bu_{n+1} + cu_n &= A\alpha^n\left(a\alpha^2 + b\alpha + c\right) + B\beta^n\left(a\beta^2 + b\beta + c\right) \\ &= A\alpha^n \cdot 0 + B\beta^n \cdot 0 = 0, \end{aligned}$$

because α and β are roots of $ax^2+bx+c=0$. So the recurrence holds for every $n \geq 0$.

With the particular A and B given, the formula also reproduces the starting values. For $n=0$,

$$A + B = \frac{(u_1 - \beta u_0) - (u_1 - \alpha u_0)}{\alpha - \beta} = \frac{(\alpha-\beta)u_0}{\alpha-\beta} = u_0,$$

and for $n=1$,

$$A\alpha + B\beta = \frac{\alpha u_1 - \alpha\beta u_0 - \beta u_1 + \alpha\beta u_0}{\alpha - \beta} = \frac{(\alpha-\beta)u_1}{\alpha-\beta} = u_1.$$

Dividing by $\alpha - \beta$ is allowed because the roots are distinct.

It is the only solution

> **Ask yourself** What exactly are we inducting on, and why do we need two terms at a time?

Suppose $u_0, u_1, u_2, \ldots$ is any sequence satisfying the recurrence, and let A, B be built from its u_0 and u_1 as above. Write $w_n = A\alpha^n + B\beta^n$. We prove by induction on n the statement

$$P(n): \quad u_n = w_n \text{ and } u_{n+1} = w_{n+1}.$$

Each new term depends on the two before it, so the statement has to carry two consecutive terms.

Base case. $P(0)$ says $u_0 = w_0$ and $u_1 = w_1$, which is what we checked above.

Inductive step. Assume $P(n)$. Both (u_n) and (w_n) satisfy the recurrence, and $a \neq 0$, so

$$u_{n+2} = -\frac{bu_{n+1} + cu_n}{a} = -\frac{bw_{n+1} + cw_n}{a} = w_{n+2}.$$

Together with $u_{n+1} = w_{n+1}$ this is $P(n+1)$.

So $P(n)$ holds for all $n \geq 0$, and $u_n = A\alpha^n + B\beta^n$ for every n. Any solution of the recurrence is of this form, with A and B fixed by u_0 and u_1. The formula gives the only solution.

The sequence v_n

Every coefficient is a product of two of the three factors $(n+1)$, $(n+2)$, $(n+3)$, each missing a different one. Dividing by $(n+1)(n+2)(n+3)$, which is never zero for $n \geq 0$, gives

$$8\,\frac{v_{n+2}}{n+3} - 2\,\frac{v_{n+1}}{n+2} - \frac{v_n}{n+1} = 0.$$

Each fraction has the form $v_k/(k+1)$. So put

$$w_n = \frac{v_n}{n+1},$$

and the recurrence becomes

$$8w_{n+2} - 2w_{n+1} - w_n = 0, \qquad w_0 = 0, \quad w_1 = \tfrac{1}{2}.$$

The auxiliary equation is $8x^2 - 2x - 1 = 0$, that is, $(2x-1)(4x+1) = 0$. Its roots $\alpha = \frac{1}{2}$ and $\beta = -\frac{1}{4}$ are real and distinct, so the first part applies, with $\alpha - \beta = \frac{3}{4}$:

$$A = \frac{w_1 - \beta w_0}{\alpha - \beta} = \frac{1/2}{3/4} = \frac{2}{3}, \qquad B = \frac{w_1 - \alpha w_0}{\beta - \alpha} = \frac{1/2}{-3/4} = -\frac{2}{3}.$$

Hence $w_n = \frac{2}{3}\left(\left(\frac{1}{2}\right)^n - \left(-\frac{1}{4}\right)^n\right)$. The first part also tells us this is the only possibility for w_n, and so for v_n. Multiplying back by $n+1$,

$$\boxed{v_n = \frac{2(n+1)}{3}\left(\frac{1}{2^n} - \left(-\frac{1}{4}\right)^n\right)} \qquad (n = 0, 1, 2, \ldots).$$

Checks

The formula gives $v_0 = \frac{2}{3}(1-1) = 0$ and $v_1 = \frac{4}{3}\left(\frac{1}{2}+\frac{1}{4}\right) = 1$, as required. Running the original recurrence by hand with $n = 0$ and $n = 1$ gives $16v_2 = 6v_1 + 6v_0$, so $v_2 = \frac{3}{8}$, and then $48v_3 = 16v_2 + 12v_1$, so $v_3 = \frac{3}{8}$. The formula agrees: $v_2 = 2\left(\frac{1}{4}-\frac{1}{16}\right) = \frac{3}{8}$ and $v_3 = \frac{8}{3}\left(\frac{1}{8}+\frac{1}{64}\right) = \frac{3}{8}$. We checked the formula exactly against the recurrence, in rational arithmetic, for n up to 41, and the first part on hundreds of random quadratics and starting values.

Common slips

- Doing the induction on one term at a time. Knowing only $u_n = w_n$ doesn't let you deduce anything about u_{n+1}: the hypothesis must cover two consecutive terms.
- Forgetting to say why we may divide by a (it is non-zero because the equation has two roots) and by $\alpha - \beta$ (the roots are distinct).
- Applying the first part to v_n directly. Its coefficients depend on n, so it is the rescaled sequence $w_n = v_n/(n+1)$ that the first part applies to.

Beyond STEP

The solutions of a second-order linear recurrence form a two-dimensional vector space, and α^n, β^n are a basis for it. This is exactly parallel to the differential equation $ay'' + by' + cy = 0$ with solutions $e^{\alpha x}$ and $e^{\beta x}$. Writing the recurrence as $\mathbf{u}_{n+1} = M\mathbf{u}_n$ for a 2×2 matrix M shows that α and β are the eigenvalues of M, and repeated roots correspond to a matrix that can't be diagonalised. Strang's *Introduction to Linear Algebra* treats difference equations through eigenvalues in just this way, and Boyce and DiPrima's *Elementary Differential Equations* develops the parallel theory for differential equations.

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